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#include <stdio.h>

#include <string.h>


#define MAX 100


char undoStack[MAX][100];
char redoStack[MAX][100];
int undoTop = -1, redoTop = -1;


// Push text state onto a stack
void push(char stack[][100], int *top, char text[]) {
    if(*top < MAX - 1) {
        (*top)++;
        strcpy(stack[*top], text);
    }
}


// Pop text from a stack
void pop(char stack[][100], int *top, char text[]) {
    if(*top >= 0) {
        strcpy(text, stack[*top]);
        (*top)--;
    }
}


int main(){
```

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char text[100] = ""; // current text
char temp[100];
int choice;

while(1) {
    printf("\n--- TEXT EDITOR (STACK Simulation: Undo / Redo) ---\n");
    printf("Current Text: \"%s\"\n", text);

    printf("1. Type New Text\n");
    printf("2. Undo\n");
    printf("3. Redo\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    getchar(); // flush newline

    if(choice == 1) {
        push(undoStack, &undoTop, text); // Save current before typing
        printf("Enter new text: ");
        fgets(text, sizeof(text), stdin);
        text[strcspn(text, "\n")] = '\0';

        redoTop = -1; // Clear redo stack after new editing

    } else if(choice == 2) {
        if(undoTop == -1) {
            printf("Nothing to UNDO!\n");
        } else {

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        push(redoStack, &redoTop, text); // Save for Redo
        pop(undoStack, &undoTop, text); // Undo
    }

} else if(choice == 3){
    if(redoTop == -1){
        printf("Nothing to REDO!\n");
    } else {
        push(undoStack, &undoTop, text); // Save for Undo
        pop(redoStack, &redoTop, text); // Redo
    }

} else if(choice == 4){
    printf("Exiting Program...\n");
    break;

} else {
    printf("Invalid choice!\n");
}

return 0;
}

```

OUTPUT

--- TEXT EDITOR (STACK Simulation: Undo / Redo) --- Current Text: ""

1. Type New Text
2. Undo

3. Redo
4. Exit Enter your choice: 1 Enter new text: HI

--- TEXT EDITOR (STACK Simulation: Undo / Redo) --- Current Text: "HI"

1. Type New Text
2. Undo
3. Redo
4. Exit Enter your choice: 2

--- TEXT EDITOR (STACK Simulation: Undo / Redo) --- Current Text: ""

1. Type New Text
2. Undo
3. Redo
4. Exit Enter your choice: 3

--- TEXT EDITOR (STACK Simulation: Undo / Redo) --- Current Text: "HI"

1. Type New Text
2. Undo
3. Redo
4. Exit Enter your choice: 4 Exiting Program...

=== Code Execution Successful ===